ORGANIC CHEMISTRY

Lab preparation (Reaction only) of:

Chloroform (trichloromethane), Nitrobenzene, Formic acid (methanoic acid), Aniline, Diethyl ether (Ethoxy ethane)

Almost sure:

I. Identification of 1, 2, and 3 alcohols by Victor Meyer's method.

II. Separation of 1, 2, and 3 amines by Hoffmann's method.

Name reactions:

1. Wurtz reaction
2. Riemer-Tiemann reaction
3. Dehydrohalogenation reaction/Elimination
4. Carbylamines reaction/Isocyanide test
5. Fittig reaction
6. Wurtz-Fittig reaction
7. Sandmeyer reaction
8. Gattermann reaction
9. Diazotization reaction
10. Dow's process
11. Friedel-Crafts reaction
12. Oxo process(Carbonylation)
13. Fermentation
14. Esterification reaction
15. Hydroboration-Oxidation reaction
16. Aldol condensation reaction
17. Perkins condensation
18. Claisen Condensation reaction
19. Cannizaro's reaction
20. Benzoin condensation
21. Aminolysis
22. Dehydration
23. Catalytic dehydrogenation
24. Coupling reaction (preparation of azo dyes)
25. Diazotization reaction
26. Friedel- Craft's reaction
27. Hoffmann's Bromamide reaction (Decarbonylation reaction)
28. Kolbe's reaction(Carboxylation reaction)
29. Rosenmund reduction
30. Williamson's etherification reaction
31. Hell-Volhard-Zelinsky reaction(HVZ)
32. Decarboxylation reaction
33. Clemensen's reduction reaction
34. Ozonolysis
35. Wolf-Kishner reduction reaction
36. Darzen's reaction

Test reaction:

- ✓ Iodoform test
- ✓ Silver mirror test (reaction with Tollen's reagent)
- ✓ Fehling's test
- ✓ Nitrous acid test of 1^0, 2^0, and 3^0 amines
- ✓ 2,4 – DNP test
- ✓ Carbylamines test (test of primary amines)
- ✓ Test of carboxylic acid
- ✓ $FeCl_3$ test
- ✓ Bromine water test
- ✓ Libermann test

How would you distinguish?

- Pentan-2-one to pentan-3-one
- Acetaldehyde and acetone
- Benzaldehyde and ethanal
- Methanal and ethanal
- Propanal and propanone
- Methanol and ethanol
- Ethanamine from N-methyl methanamine

- ethanoic acid (acetic acid) from methanoic acid (formic acid)
- Acid present in venom of bee and acid present in vinegar
- Ethanol and acetic acid

Important reactions:

Reduction of nitrobenzene in different medium, Preparation of alcohols using Grignard reagent, Oxidation of alcohols, Fermentation, All reactions of chloroform.

Reactions to prepare: (with one use)

DDT, Picric acid, BHC, Urotropine, Chloropicrin, TNT

Convert:

- Methane to ethane
- Ethane to ethene
- Methane to ethanoic acid
- Ethanol to ethyne
- Ethane to propanamine
- 1-bromopropane to 2-bromopropane and vice-versa
- Benzene to benzoic acid
- Chlorobenzene to aniline
- Aniline to benzyl amine
- Nitrobenzene to bromobenzene
- Propan-1-ol to propan-2-ol and vice versa.
- Methanol to ethanol and vice versa
- n-propyl alcohol to iso-propyl alcohol
- ethanol into acetylene
- ethanol to propanoic acid
- Ethyl alcohol to tert-butyl alcohol.
- Sucrose to ethanol
- Ethyl alcohol to acetone
- 1-bromopropane to 2- bromopropane and vice-versa.
- Propanol to 2- propanol and vice versa.
- Methanamine to ethanamine and vice versa.
- Phenol to anisole (methoxy benzene) and vice versa.
- Ethoxy ethane to methoxy ethane.
- ethoxyethane to ethanoyl chloride.
- Phenol/aniline to azo-dye.
- Phenol into picric acid
- Phenol into Phenolpthalein
- Ethanal/ ethanol to 3-hydroxy butanal.
- Ethanol to 2- hydroxy propanoic acid.
- Propanone (acetone) to 2-hydroxy-2-methyl propanoic acid.
- Phenol to toluene.
- Ethanol to propanol/ propanoic acid.
- Acetylene to acetic acid
- Toluene to m-nitrobenzoic acid
- Ethanol to acetone
- Acetaldehyde to acetone
- Phenol to benzaldehyde
- Acetaldehyde to acetone and vice-versa
- Methanol to ethanal
- Ethanol to methanol
- Propanone into 2-hydroxy-2-methyl propanoic acid
- Acetylene to acetic acid
- Peopanone to methanol
- Propanone to 4-hydroxy-4-methyl pentan-2-one
- Acetaldehyde to lactic acid
- Benzene to benzoic acid.
- Aniline to benzoic acid.
- Methanoic acid to ethanoic acid and vice-versa.
- Benzonitrile to m-nitrobenzoic acid.
- Phenol to bemzoic acid and vice-versa.
- Ethanoyl chloride to methanol.
- Benzamide to toluene.
- Ethyl acetate to acetoacetic ester.
- Methanamide to ethanamine.

Some Important Questions of Organic Chemistry:

- Why is chloroform stored in a dark bottle containing ethanol?
- Why does chloroform not give white ppt. with aq. $AgNO_3$?
- A small amount of ethanol is added in the bottle of chloroform.
- Ethanol is soluble in water but ethyl chloride is not soluble.
- Boiling point of n-butyl alcohol is higher than that of tert-butyl alcohol.
- Chloroform is not used as an anesthetic these days.
- Alkyl halide gives alcohol with aq. KOH while with alcoholic KOH gives alkene.
- Haloalkane reacts with KCN to give alkyl cyanide but with AgCN gives alkyl isocyanide
- Melting point of p-dichlorobenzene is higher than that of o- and m-dichlorobenzene.
- Why is nucleophilic substitution reaction difficult in haloarenes?
- Why is chlorobenzene o/p – directing towards the electrophilic substitution reaction?
- Oxidation of a tertiary alcohol occurs only in drastic conditions.
- Ethanol is more acidic than propan-2-ol.
- Lower alcohols are soluble in water, whereas higher alcohols are insoluble in water.
- Alcohols are not dehydrated by $CaCl_2$.
- Alcohol is a weaker acid than water.
- Ethanol is more acidic than propan-2-ol.
- Why is the boiling point of ethanol greater than that of ethoxy ethane?
- Why is phenol more acidic than aliphatic alcohol?
- Why is phenol o/p – directing towards the electrophilic substitution reaction?
- Why does nitrobenzene undergo an electrophilic substitution reaction at Meta position? (Explain why $-NO_2$ group is meta-directing and ring deactivating towards electrophilic aromatic substitution)
- It is dangerous to boil a sample of ether stored for a long time. Give a reason.
- Ether is stored in a bottle containing an iron wire. Why?
- Anisole is prepared by using sodium phenolate and methyl iodide but not sodium methoxide and chlorobenzene.
- Why is chloroacetic acid a stronger acid than acetic acid?
- Why is formic acid a stronger acid than acetic acid?
- Dichloroacetic acid is a stronger acid than monochloroacetic acid.
- Why nitro group is called an ambident group?
- Why amino group of aniline is protected before nitration? (Aniline cannot be nitrated directly, why?)

Some Important Questions of Organic Chemistry:

- Write the functional isomers of C_3H_6O and $C_4H_{10}O$with their IUPAC name. Give a chemical test to distinguish them.
- Write an unsymmetrical ether of C_3H_8O. How would you prepare this ether using Williamson's synthesis?
- Write down possible isomeric amines of C_3H_9N and give their IUPAC names. Give basicity order of these isomers.

What happens when?

- Propan-2-ol is treated with sodium.
- Ethanol is dehydrated?

- Vapour of propan-2-ol is passed over heated Cu at 300^0
- Molasses is fermented in the presence of yeast.
- Ethanoic acid is heated with methanol in the presence of conc. H_2SO_4
- Sodium benzoate is heated with sodalime.
- Phenol is heated with zinc dust.
- Chlorobenzene is treated with chloral.
- Ethyl alcohol (ethanol) is treated with acetic acid (ethanoic acid)
- Phenol is treated with aq. Br_2.
- Aniline is treated with aq. Br_2.
- Phenol is treated with benzene diazonium chloride.
- Ethoxy ethane is treated with excess HI.
- Methanal(formaldehyde) is treated with ammonia.
- Methanal/benzaldehyde is treated with NaOH.
- Ethanal/propanone is treated with NaOH.
- Aldehyde/ketone is treated with hydroxyl amine.
- Ethanol is heated with conc. H_2SO_4.
- Acetic acid is treated with P_2O_5.
- Methanoic acid is heated
- Tollen's reagent is warmed with methanoic acid.
- Acetic acid is treated with sodium metal.
- Benzoic acid is treated with PCl_5.
- Benzoic acid is treated with phenol in acidic medium.
- Benzoyl chloride is treated with H_2 in presence of Pd and $BaSO_4$.
- Ethanoic acid is heated with HI in presence of red phosphorous.
- Acetyl chloride reacts with phenol in presence of anhydrous $AlCl_3$.
- Ethanoyl chloride and ammonia is heated with Br_2 and aqueous KOH.
- Ethanoic anhydride is reduced.
- Ethanamide is reduced.

PHYSICAL CHEMISTRY

Volumetric Analysis

1. Define Molarity, Normality, normal solution, decinormal solution, molar solution, decimolar solution, molality, ppm, ppb, titration, indicator, equivalence point, end point, normality factor, titration error.

2. Derive $N_1V_1 = N_2V_2$ or $S_1V_1 = S_2V_2$

3. Derive the relation between normality and molarity.

4. Write differences between end point and equivalence point, primary and secondary standard substance/ solution, titrant and titrand.

5. Numerical problems

a) Dilution: $N_1V_1 = N_2V_2$

b) Mixture: $NmVm = N_1V_1 + N_2V_2$

c) Acid-base mixture: NmVm = NaVa - NbVb

d) Solution solid neutralization: W/E = NV/1000

e) Metal, acid and base neutralization

f) Without dilution NaVa= W/E + NbVb

g) With dilution First find the volume of base required, then NaVa= W/E + NbVb

h) W/E= N_1V_1 - N_2V_2 /1000

i) Equivalent wt. of $KMnO_4$ in acidic, basic and neutral medium.

Ionic Equilibrium

1. State and explain Ostwald's dilution law. Mention its significance. What is its limitation?

2. Define degree of ionization. Mention the factors affecting degree of ionization of weak electrolytes.

3. What is the effect of temperature and common ion on degree of ionization of weak electrolytes?

7. Show your acquaintance with a) Arrhenius concept of acids and bases. b) Bronsted-Lowry concept of acids and bases. c) Lewis concept of acids and bases.

8. What do you mean by conjugate acid –base pairs? Write the conjugate acid and base each of

i) H_2O ii) NH_3 iii) HCO_3^-

9. Discuss about the relative strength of conjugate acid- base pairs.

10. Define ionic product of water. Mention its significance. How does temperature effect Kw values?

11. Define pH and pOH. Prove that:

$$pH + pOH = 14$$

12. Define common ion effect. Give an example. Write its two applications.

13. What is solubility product principle? Discuss the application of solubility product principle and common ion effect in the qualitative analysis of basic radicals.

14. Define salt hydrolysis. On the basis of this concept discuss the nature of sodium carbonate and copper sulphate solution.

15. What is a buffer solution? Give an example of an acidic and basic buffer each.

16. What do you mean by titration curve? Discuss the selection of a suitable indicator in acid-base titration.

Numerical

1. Calculate the pH of 0.2N H_2SO_4 solution. [0.6]

2. Calculate the pH of 1×10^{-8} M HCl solution. [6.95]

3. 200 ml of an aqueous solution of HCl having pH 2 is mixed with 300 ml of an aqueous solution of NaOH having pH 12. What will be the pH of the resulting mixture? [11.3]

4. Equal volume of two solutions having pH 4 and pH 5 respectively are mixed together. Calculate the pH of the resulting mixture. [4.25]

5. Calculate the pH of 0.1 molar solution of Acetic Acid. Given that the ionization constant of Acetic acid is (Ka) $=1.8\times10^{-5}$. [2.87]

6. The solubility product of chalk (Calcium Carbonate) is 9.3×10^{-8}. Calculate its solubility in gram per litre. [Ans: $3.05\text{x}10^{-2}$]

7. The solubility product of silver chloride (AgCl) is 1×10^{-10} at 298K. Calculate its solubility in

a) Pure water [Ans: 10^{-5} M]

b) 0.02M KCl solution. [$5\text{x}10^{-9}$ M]

c) Also calculate the amount of silver chloride precipitated in KCl solution. [$1.4\text{x}10^{-3}$ g/L]

8. The solubility product of calcium hydroxide [$Ca(OH)_2$] at 25 ^{0}C is 4.42×10^{-5}. A 500 ml of a saturated solution of calcium hydroxide is mixed with an equal volume of 0.4 M solution of NaOH. How much calcium hydroxide is precipitated in the mixture? [0.74 g]

9. 0.05 g of $AgNO_3$ is added to 500 mL of 0.001 M NaCl solution. Predict whether precipitation occurs or not? If occurs, calculate the mass of AgCl precipitated. [Ksp of AgCl= 1.8×10^{-10}]
Ans: 0.04 g

Chemical Kinetics

1. Define: rate of reaction, order of reaction, molecularity, rate law, rate constant, activation energy, effective collision, and half-life of a reaction.

2. Write the differences between i) rate of reaction and rate constant ii) order and molecularity

3. Explain: i) Factors affecting rate of reaction ii) Collision theory of reaction rates

4. Derive i) Integrated rate law for first order reaction and its half-life. ii) Integrated rate law for zero order reaction and its half-life.

5. Draw energy profile diagram for catalyzed and uncatalyzed reactions.

6. Numerical involving rate expression and equivalent rate.

7. Numerical involving integrated rate law and half-life.

8. Numerical involving initial rate method (tabular numerical)

9. Write the Arrhenius equation and mention its significance.

Thermodynamics

1. Define: state of a system, state function, extensive and intensive properties, internal energy, enthalpy, entropy, Gibb's free energy, enthalpy of formation, standard enthalpy of combustion. 2. State and illustrate Hess's law. Write its importance.

3. Explain entropy criteria of spontaneity.

4. Discuss enthalpy criteria of spontaneity.

5. State and explain first law and second law of thermodynamics.

6. Derive Gibb's Helmholtz equation. Describe the conditions for spontaneous and non-s spontaneous chemical reactions on the basis of it.

7. What is the effect of temperature in feasibility of endothermic and exothermic reactions? Explain in the light of Gibb's Helmholtz equation.

8. Prove that useful work done by a system is equal to the decrease in free energy of the system.
9. Numerical involving Hess's law/ enthalpy of formation and Combustion.

10. Numerical involving $\Delta G = \Delta H - T\Delta S$

Electrochemistry

1. Define: Galvanic cell, single electrode potential, standard electrode potential, electrochemical series, and standard cell potential

2. What is a salt bridge? Write its functions.

3. What are primary and secondary cells? Give examples.

4. What is a fuel cell? What is its significance?

5. What is a standard hydrogen electrode? Why is it used? What are its limitations?

6. Name the secondary reference electrode with its cell notation.

7. Derive the relation between standard cell potential and free energy change of the cell

reaction. 8. Calculate the standard EMF of a cell which involves the following cell reaction

$Zn + 2Ag^{+} \rightarrow Zn^{2+} + 2Ag$

Given that $E°Zn^{++}/Zn = – 0.76$ V and $E°Ag^{+}/Ag = + 0.80$ V

9. You are given the standard reduction electrode potential of the Ni^{2+}/ Ni and Cu^{2+}/Cu as –0.25 V and +0.34V, respectively.

(a) Construct a galvanic cell indicating cathode and anode.

(b) Write the cell reaction.

(c) Calculate the EMF at 1 M solution of ions.

d) This above reaction is spontaneous or not?

(e) Will the reaction Cu(s) + Ni++ (aq) → Cu++ (aq) + Ni(s) occur?

10. Can a solution of 1 M copper sulphate be stored in a vessel made of nickel? Given that $E°Ni^{2+}/Ni = – 0.25$ V; $E°Cu^{2+}/ Cu = + 0.34$ V

11. Given the standard reduction potential as, $E^{0}Mg^{++}/Mg = –2.37$ V; $E^{0}Fe^{2+}/Fe = –0.44$ V $E^{0}2H^{+}/H^{2} = 0.00$V and $E^{0}Ag^{+}/Ag = +0.80$ V

(a) Which one acts as strong reducing agent and why?

(b) Which one acts as strong oxidizing agent and why?

(c) Compare the reactivity of the metals.

(d) Which one cannot liberate hydrogen gas on treating with dilute acids and why?

12. Given standard reduction potential of Fe^{2+}/Fe and Pb^{2+}/Pb are -0.44 V and -0.13 V respectively.

i) Write O. half and R. half reactions

ii) Write the net cell reaction

iii) Calculate emf of the cell

iv) Calculate standard Gibb's free energy change of the reaction.

INORGANIC CHEMISTRY

Transition elements

1. Define: Transition elements, complex, ligand, coordination number

2. Give reasons: Transition elements show

- Variable oxidation state
- Formation of complexes
- Reason for colored complexes
- Catalytic property
- Magnetic property.
- CFT for octahedral complex.

3. Write: Characteristics of transition elements, examples of tetrahedral, square planar and octahedral complexes.

4. Explain: Crystal field splitting in octahedral complexes, d-d transition

5. Cu^{2+} complexes are colored while those of Zn^{2+} are colorless. Explain the reason.

6. Why is zinc not considered as transition element?

Heavy metals

1. Name: Chief Ores of copper, iron, zinc, mercury and silver

2. Explain:

i. Extraction of copper, iron, zinc and mercury

ii. Manufacture of steel by open hearth and basic oxygen process

Iii.Chemistry of blue vitriol, white vitriol, calomel, and corrosive sublimate

Iv. Rusting (corrosion) of iron and methods of prevention.

v.What is Rinman's green? How it is prepared? Write its one important use.

vi.What is lithopone? How it is prepared?

vii. Write molecular formula of Philosopher's wool. How it is prepared?

Viii.What is Nessler's reagent? How is it prepared? Give its one use.

ix. What is lunar caustic? Why it is called lunar caustic?

APPLIED CHEMISTRY

1. Define: Homopolymer, copolymer, auxochrome, chromophore, cement, clinker, paper, pulp, radioactivity, nuclear transmutation

2. Write monomers and one use of:

(a) Bakelite, (b) Nylon-6, 6, (c) Polyvinyl chloride (PVC) (d) Teflon (e) Polystyrene

3. Write an example and one use of:

i. Direct dye, vat dye, and mordant dye

ii. Analgesic, antipyretic, tranquilizer, and antiseptic drug.

iii. Pesticide- insecticides, herbicides, fungicides

iv. Nuclear fusion and nuclear fission

4. Differentiate between

i. OPC and PPC

ii. Hydraulic and non-hydraulic cement

iii. Nuclear fission and nuclear fusion

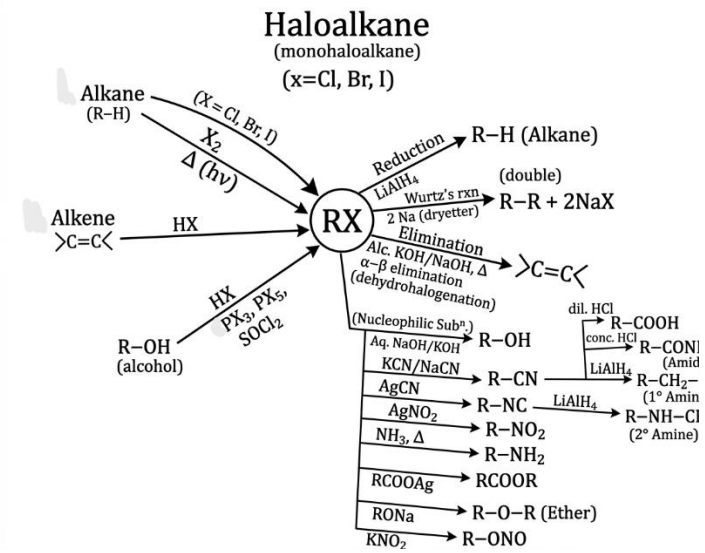
iv. Artificial and natural radioactivity

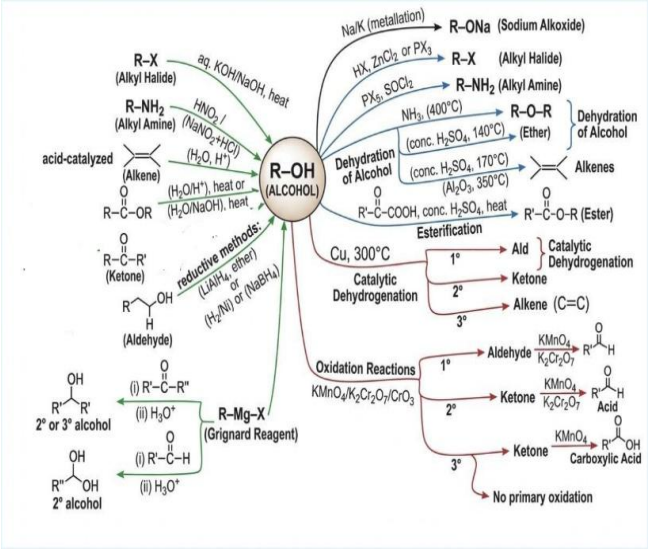
v. Paper and pulp

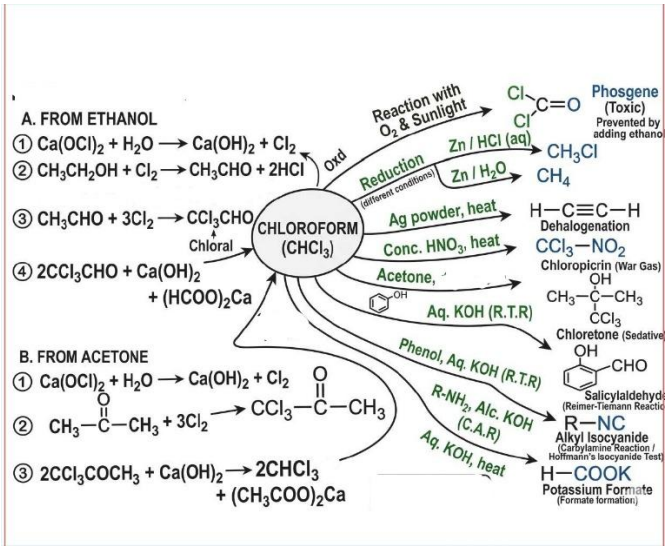
vii. Nuclear and chemical reaction

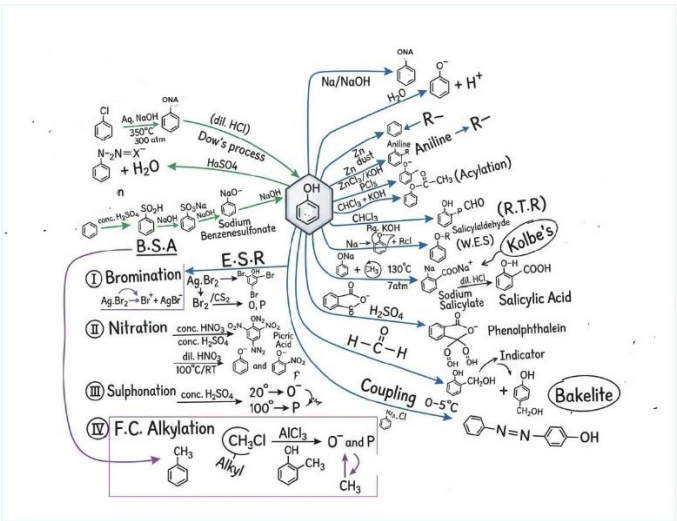
5. Draw: Flow sheet diagram of i) cement production and ii) paper production

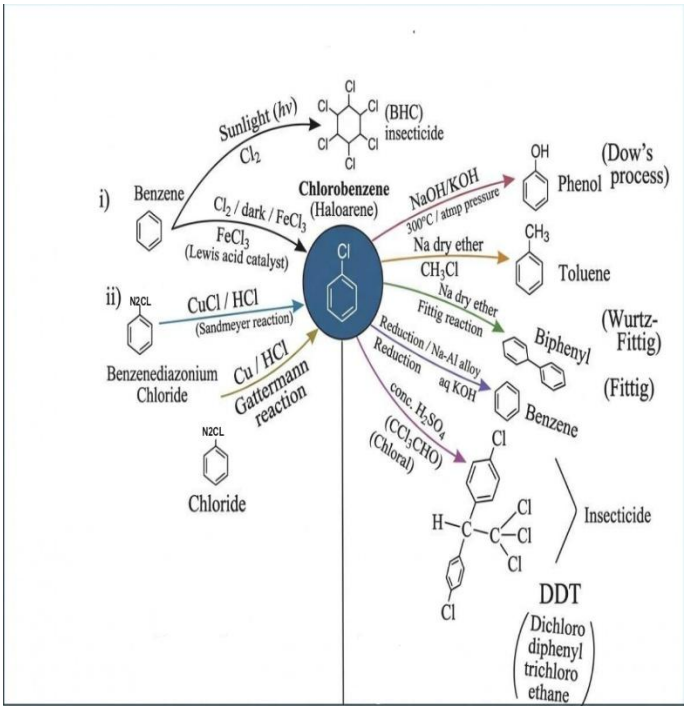
6. Unit of Radioactivity.

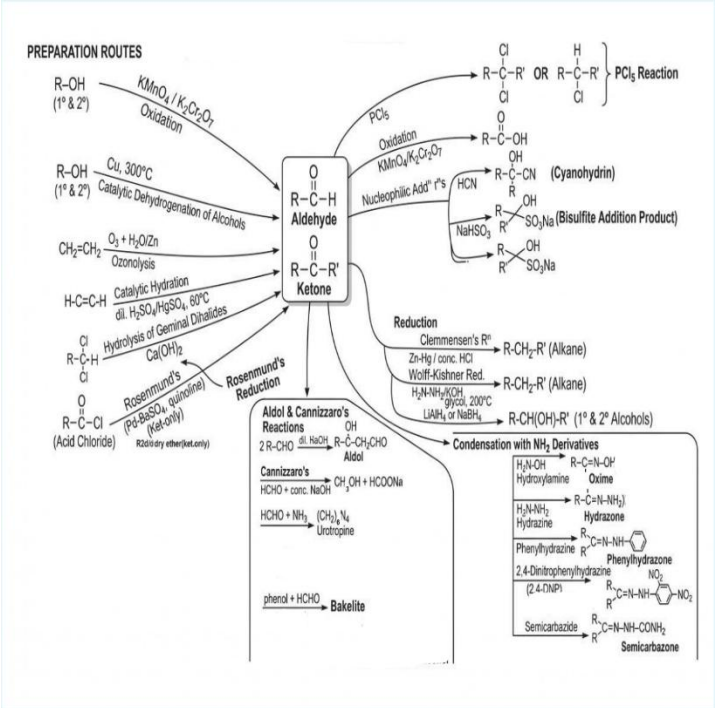

Haloalkane
(monohaloalkane)
(x=Cl, Br, I)
Alkane
(R–H)
(X = Cl, Br, I)
X2
Δ (hν)
Alkene
>C=C<
HX
R–OH
(alcohol)
HX
PX3, PX5,
SOCl2
RX
Reduction
LiAlH4
R–H (Alkane)
Wurtz's rxn
2 Na (dryetter)
(double)
R–R + 2NaX
Elimination
Alc. KOH/NaOH, Δ
α–β elimination
(dehydrohalogenation)
>C=C<
(Nucleophilic Subⁿ)
Aq. NaOH/KOH
R–OH
KCN/NaCN
R–CN
AgCN
AgNO2
R–NC
NH3, Δ
R–NH2
RCOOAg
RCOOR
RONa
R–O–R (Ether)
KNO2
R–ONO
dil. HCl
R–COOH
conc. HCl
R–CONH
(Amid
LiAlH4
R–CH2–
(1° Amin
LiAlH4
R–NH–Cl
(2° Amine)

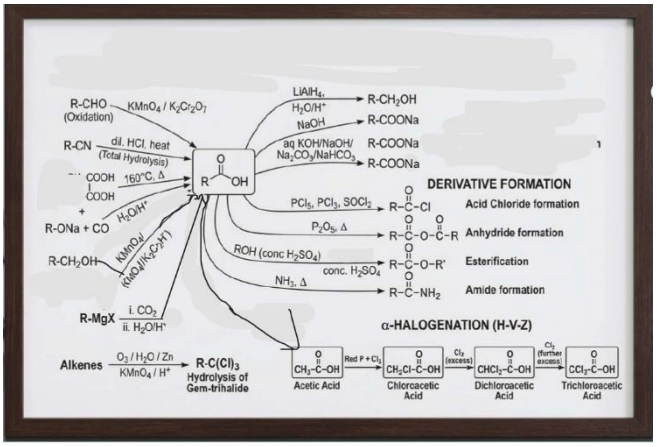

R–X
(Alkyl Halide)
aq. KOH/NaOH, heat
R–NH2
(Alkyl Amine)
HNO2 /
(NaNO2+HCl)
acid-catalyzed
(Alkene)
(H2O, H+)
(H2O/H+), heat or
(H2O/NaOH), heat
(Ketone)
(Aldehyde)
reductive methods:
(LiAlH4, ether)
or
(H2/Ni) or (NaBH4)
R–Mg–X
(Grignard Reagent)
2° or 3° alcohol
2° alcohol
R–OH
(ALCOHOL)
NaK (metallation)
R–ONa (Sodium Alkoxide)
HX, ZnCl2 or PX3
R–X (Alkyl Halide)
PX5, SOCl2
R–NH2 (Alkyl Amine)
Dehydration of Alcohol
NH3, (400°C)
R–O–R
(Ether)
Dehydration of Alcohol
(conc. H2SO4, 140°C)
(conc. H2SO4, 170°C)
(Al2O3, 350°C)
Alkenes
R'–C–COOH, conc. H2SO4, heat
R'–C–O–R (Ester)
Esterification
Cu, 300°C
Catalytic Dehydrogenation
1°
Ald
Catalytic Dehydrogenation
2°
Ketone
3°
Alkene (C=C)
Oxidation Reactions
KMnO4/K2Cr2O7/CrO3
1°
Aldehyde
KMnO4
K2Cr2O7
2°
Ketone
KMnO4
K2Cr2O7
Acid
3°
Ketone
KMnO4
Carboxylic Acid
No primary oxidation

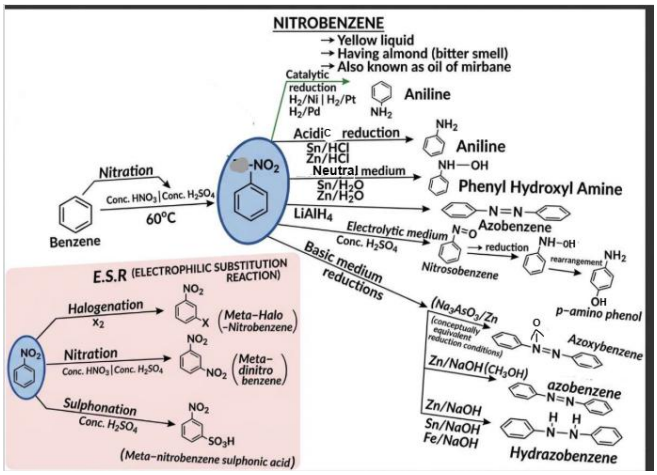

A. FROM ETHANOL
① Ca(OCl)2 + H2O → Ca(OH)2 + Cl2
② CH3CH2OH + Cl2 → CH3CHO + 2HCl
Oxd
③ CH3CHO + 3Cl2 → CCl3CHO
Chloral
④ 2CCl3CHO + Ca(OH)2
+ (HCOO)2Ca
CHLOROFORM
(CHCl3)
Reaction with
O2 & Sunlight
Phosgene
(Toxic)
Prevented by
adding ethanol
Reduction
(different conditions)
Zn / HCl (aq)
CH3Cl
Zn / H2O
CH4
Ag powder, heat
H–C≡C–H
Dehalogenation
Conc. HNO3, heat
CCl3–NO2
Chloropicrin (War Gas)
Acetone,
Aq. KOH (R.T.R)
Chloretone (Sedative)
Phenol, Aq. KOH (R.T.R)
Salicylaldehyde
(Reimer-Tiemann Reactio
R-NH2, Alc. KOH
(C.A.R)
R–NC
Alkyl Isocyanide
(Carbylamine Reactio /
Hoffmann's Isocyanide Test)
Aq. KOH, heat
H–COOK
Potassium Formate
(Formate formation)
B. FROM ACETONE
① Ca(OCl)2 + H2O → Ca(OH)2 + Cl2
② CH3–C–CH3 + 3Cl2 → CCl3–C–CH3
③ 2CCl3COCH3 + Ca(OH)2 → 2CHCl3
+ (CH3COO)2Ca

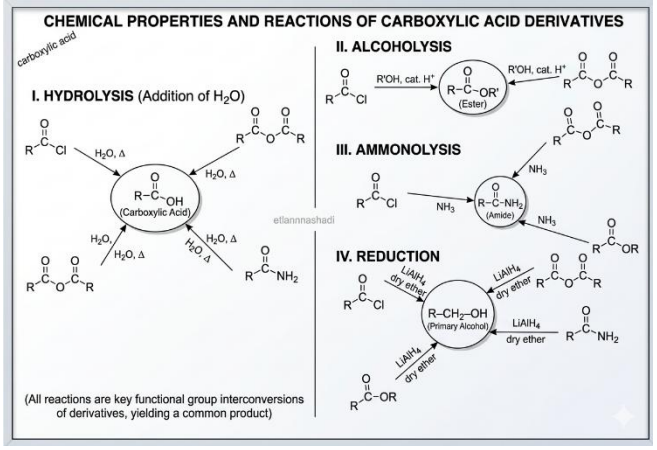

Aq. NaOH
350°C
300 atm
(dil. HCl)
Dow's process
HaSO4
Na/NaOH
H2O
+ H+
R–
Aniline
Zn dust
ZnCl/KOH
PCl5
(Acylation)
CHCl3 + KOH
Sodium Benzenesulfonate
B.S.A
E.S.R
Salicylaldehyde
(R.T.R)
(W.E.S)
Kolbe's
Sodium Salicylate
Salicylic Acid
Bromination
Ag.Br2
Br2 /CS2
0, P
Nitration
conc. HNO3
conc. H2SO4
dil. HNO3
100°C/RT
Picric Acid
H2SO4
Phenolphthalein
Indicator
Sulphonation
conc. H2SO4
20° → O–
100° → P
Coupling
0–5°C
Bakelite
F.C. Alkylation
CH3Cl
AlCl3
O– and P
Alkyl

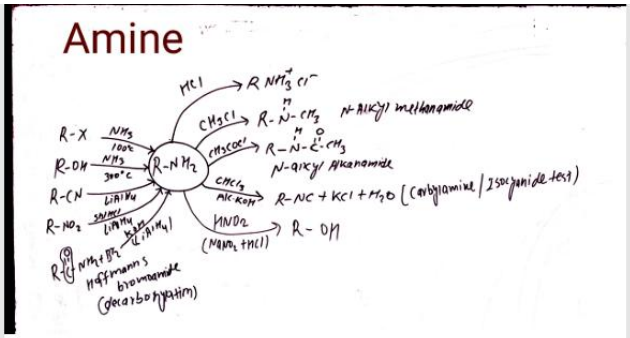

Sunlight (hν)
Cl2
(BHC)
insecticide
i)
Benzene
Cl2 / dark / FeCl3
FeCl3
(Lewis acid catalyst)
Chlorobenzene
(Haloarene)
NaOH/KOH
300°C / atmp pressure
Phenol
(Dow's process)
Na dry ether
CH3Cl
Toluene
Na dry ether
Fittig reaction
Biphenyl
(Wurtz-Fittig)
Reduction / Na-Al alloy
Reduction
aq KOH
Benzene
(Fittig)
ii)
CuCl / HCl
(Sandmeyer reaction)
Benzenediazonium Chloride
Cu / HCl
Gattermann reaction
Chloride
conc. H2SO4
(CCl3CHO)
(Chloral)
Insecticide
DDT
Dichloro diphenyl trichloro ethane

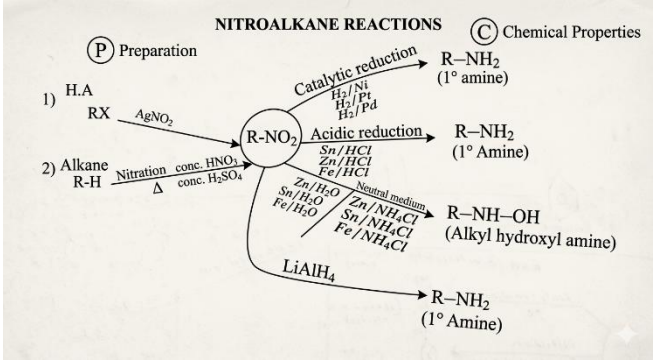

PREPARATION ROUTES
R–OH
(1° & 2°)
KMnO4 / K2Cr2O7
Oxidation
R–OH
(1° & 2°)
Cu, 300°C
Catalytic Dehydrogenation of Alcohols
CH2=CH2
O3 + H2O/Zn
Ozonolysis
H–C=C–H
Catalytic Hydration
dil. H2SO4/HgSO4, 60°C
Hydrolysis of Geminal Dihalides
Ca(OH)2
Rosenmund's
(Pd-BaSO4, quinoline)
(Ket-only)
(Acid Chloride)
Aldehyde
Ketone
Rosenmund's Reduction
PCl5
Oxidation
KMnO4/K2Cr2O7
Nucleophilic Addⁿ i"s
HCN
(Cyanohydrin)
NaHSO3
(Bisulfite Addition Product)
PCl5 Reaction
Reduction
Clemmensen's Rⁿ
R-CH2-R' (Alkane)
Zn-Hg / conc. HCl
Wolff-Kishner Red.
R-CH2-R' (Alkane)
LiAlH4 or NaBH4
R-CH(OH)-R' (1° & 2° Alcohols)
Aldol & Cannizzaro's Reactions
2 R-CHO
dil. NaOH
Aldol
Cannizzaro's
HCHO + conc. NaOH
CH3OH + HCOONa
HCHO + NH3
Urotropine
phenol + HCHO
Bakelite
Condensation with NH2 Derivatives
Hydroxylamine
Oxime
Hydrazine
Hydrazone
Phenylhydrazine
Phenylhydrazone
2,4-Dinitrophenylhydrazine
(2,4-DNP)
Semicarbazide
Semicarbazone

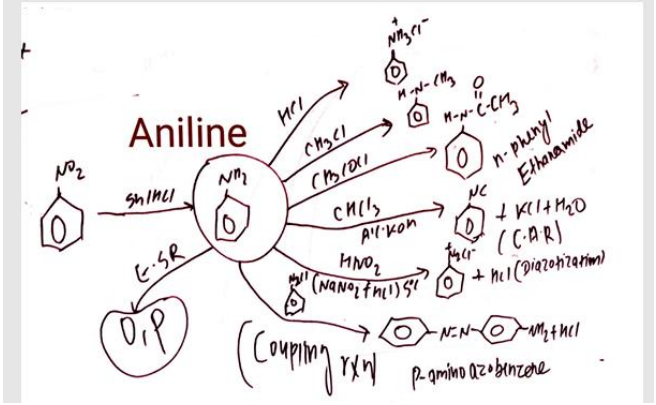

R-CHO (Oxidation)
KMnO4 / K2Cr2O7
R-CN
dil. HCl, heat (Total Hydrolysis)
160°C, Δ
R-ONa + CO
H2O/H+
R-CH2OH
KMnO4 (KMnO4/K2Cr2O7)
R-MgX
i. CO2
ii. H2O/H+
Alkenes
O3 / H2O / Zn
KMnO4 / H+
R-C(Cl)3
Hydrolysis of Gem-trihalide
LiAlH4, H2O/H+
R-CH2OH
NaOH
R-COONa
aq KOH/NaOH/ Na2CO3/NaHCO3
R-COONa
R-COONa
DERIVATIVE FORMATION
PCl5, PCl3, SOCl2
Acid Chloride formation
P2O5, Δ
Anhydride formation
ROH (conc H2SO4)
conc. H2SO4
Esterification
NH3, Δ
Amide formation
α-HALOGENATION (H-V-Z)
Acetic Acid
Red P + Cl2
Chloracetic Acid
Cl2 (excess)
Dichloracetic Acid
Cl2 (further excess)
Trichloroacetic Acid

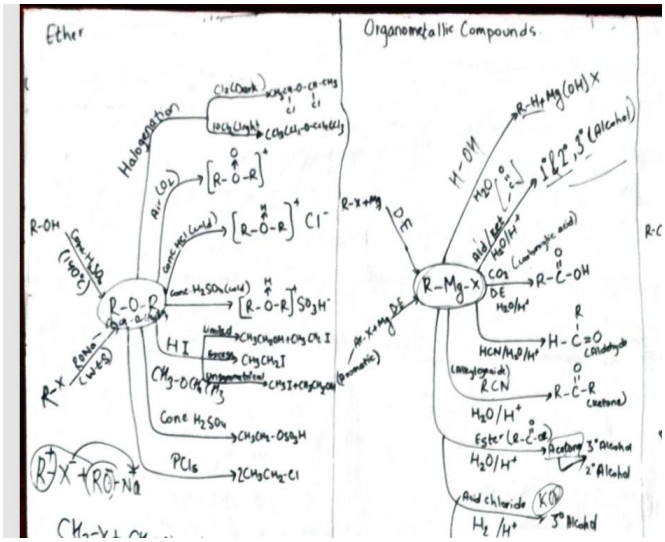

NITROBENZENE
→ Yellow liquid
→ Having almond (bitter smell)
→ Also known as oil of mirbane
Catalytic reduction
H2/Ni | H2/Pt
H2/Pd
Aniline
Acidic reduction
Sn/HCl
Zn/HCl
Aniline
Neutral medium
Sn/H2O
Zn/H2O
Phenyl Hydroxyl Amine
LiAlH4
Azobenzene
Nitration
Conc. HNO3 | Conc. H2SO4
60°C
Benzene
Electrolytic medium
Conc. H2SO4
Nitrosobenzene
reduction
rearrangement
p-amino phenol
Basic medium reductions
(Na3AsO3/Zn
Azoxybenzene
Zn/NaOH(CH3OH)
azobenzene
Zn/NaOH
Sn/NaOH
Fe/NaOH
Hydrazobenzene
E.S.R (ELECTROPHILIC SUBSTITUTION REACTION)
Halogenation
X2
(Meta-Halo -Nitrobenzene)
Nitration
Conc. HNO3 | Conc. H2SO4
(Meta-dinitro benzene)
Sulphonation
Conc. H2SO4
(Meta-nitrobenzene sulphonic acid)


CHEMICAL PROPERTIES AND REACTIONS OF CARBOXYLIC ACID DERIVATIVES
carboxylic acid
I. HYDROLYSIS (Addition of H2O)
(Carboxylic Acid)
II. ALCOHOLYSIS
R'OH, cat. H+
(Ester)
III. AMMONOLYSIS
NH3
(Amide)
IV. REDUCTION
LiAlH4 dry ether
(Primary Alcohol)
etlannnashadi
(All reactions are key functional group interconversions of derivatives, yielding a common product)


Amine
HCl
CH3Cl
N-Alkyl methanamide
CH3COCl
N-alkyl Alkanamide
CHCl3
KOH
R-NC + KCl + H2O (Carbylamine / Isocyanide test)
HNO2
(NaNO2 + HCl)
R-OH
Hoffmann's bromamide (decarbonylation)


NITROALKANE REACTIONS
P Preparation
C Chemical Properties
1) H.A
RX
AgNO2
2) Alkane
R-H
Nitration
conc. HNO3
conc. H2SO4
Δ
R-NO2
Catalytic reduction
H2/Ni
H2/Pt
H2/Pd
R–NH2
(1° amine)
Acidic reduction
Sn/HCl
Zn/HCl
Fe/HCl
R–NH2
(1° Amine)
Neutral medium
Zn/H2O
Sn/H2O
Fe/H2O
Zn/NH4Cl
Sn/NH4Cl
Fe/NH4Cl
R–NH–OH
(Alkyl hydroxyl amine)
LiAlH4
R–NH2
(1° Amine)


Aniline
HCl
CH3Cl
CH3COCl
n-phenyl Ethanamide
CHCl3
KOH
+ KCl + H2O (C.A.R)
HNO2
(NaNO2 + HCl)
+ HCl (Diazotization)
Sn/HCl
O, P
(Coupling rxn)
p-amino azobenzene


Ether
Organometallic Compounds
Halogenation
R-OH
R-O-R
HI
Conc H2SO4
PCl5
R-Mg-X
CO2
H2O/H+
HCN/H2O/H+
RCN
Ester
Acid chloride
H2/H+